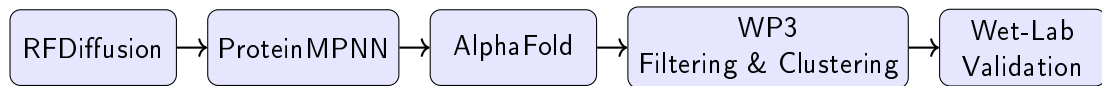


## WP3: Filtering and Clustering of AI-Generated Enzyme Mimics

- Modern AI-based enzyme design pipelines can generate thousands of enzyme candidates.
- Experimental validation is expensive and time-consuming.
- Most generated structures exhibit limited structural diversity (structural bias of ALphaFold noticed).
- **Goal of WP3:**
  - Preserve catalytic functionality;
  - Maximize structural diversity;
  - Select only the most promising candidates for laboratory validation.

# Position of WP3 within the Workflow



Generate → Sequence Design → Structure Prediction → Candidate Selection → Experimental Validation

**Different points in the workflow where to integrate Filtering & Clustering.**

## Main Objective

Explore the structural space of enzyme mimics while preserving catalytic functionality.

Two complementary strategies are employed:

### ① Filtering

- Improve candidate quality — Remove unlikely functional structures.

### ② Clustering

- Promote structural diversity.
- Ensure broad exploration of enzyme structural space.

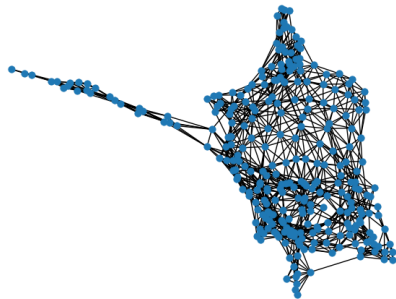
# Protein Structure as a Graph

- Each amino-acid residue becomes a graph vertex.
- Residues are connected if their  $C_\alpha$  atoms are spatially close.
- Threshold:

$$\tau = 8\text{\AA} \text{ (ångström unit)}$$

- Produces a residue interaction graph.

$$(v_i, v_j) \in E \iff \|x_i - x_j\|_2 \leq 8\text{\AA}$$



# Graph-Based Structural Features

## Basic Features

- Number of nodes
- Number of edges
- Density
- Average degree
- Minimum degree
- Maximum degree

## Connectivity Features

- Connected components
- Diameter
- Average shortest path

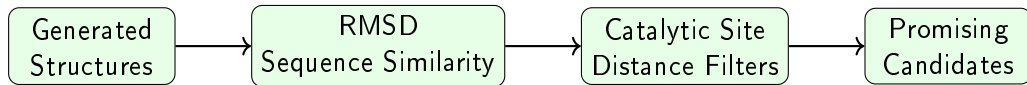
## Topological Features

- Clustering coefficient
- Degree centrality

## Advanced Features (some not yet explored)

- Catalytic residue geometry
- Minimum Dominating Set size
- Protein language model embeddings
- Structural descriptors

# Filtering Pipeline



- Remove structurally implausible candidates.
- Preserve catalytic residue organization.
- Focus computational resources on promising structures.

## Catalytic triad of turboPETase

*SER – HIS – ASP*

### Filter 1

- Compare catalytic residue positions between generated and reference enzymes.

### Filter 2

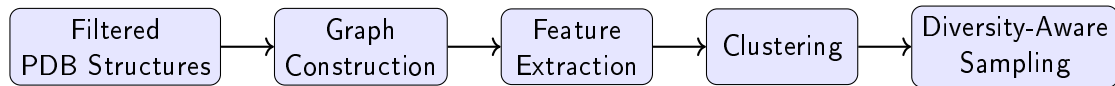
- Measure internal distances:
  - SER–HIS
  - HIS–ASP

### Objective

- Preserve catalytic geometry.
- Increase likelihood of catalytic functionality.



# Clustering Pipeline



- 1 Convert structures into graphs.
- 2 Compute structural descriptors.
- 3 Build feature space.
- 4 Identify structural families.
- 5 Select representatives from each cluster.

# Clustering Methods

- K-Means
- Hierarchical Clustering
- Spectral Clustering
- Density-Based Clustering

## Cluster quality metrics

- Silhouette score
- Intra-cluster similarity
- Inter-cluster separation

## Expected Outcome

- Distinct structural families of enzyme mimics.

# Feature Space Representation

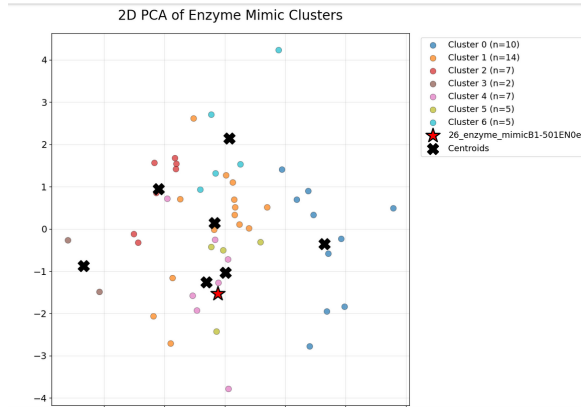
|    |                             |     |      |          |           |   |    |          |   |          |
|----|-----------------------------|-----|------|----------|-----------|---|----|----------|---|----------|
| 0  | 6_enzyme_mimicB1-501EN379c  | 308 | 1591 | 0.033652 | 10.331169 | 4 | 19 | 0.533014 | 1 | 0.033652 |
| 1  | 11_enzyme_mimicB1-501EN53d  | 324 | 1632 | 0.031189 | 10.074074 | 4 | 18 | 0.547730 | 1 | 0.031189 |
| 2  | 14_enzyme_mimicB1-501EN28d  | 335 | 1659 | 0.029654 | 9.904478  | 4 | 19 | 0.544153 | 1 | 0.029654 |
| 3  | 8_enzyme_mimicB1-501EN53a   | 325 | 1599 | 0.030370 | 9.840000  | 2 | 19 | 0.549844 | 1 | 0.030370 |
| 4  | 42_enzyme_mimicB1-501EN349a | 319 | 1607 | 0.031683 | 10.075235 | 3 | 19 | 0.537129 | 1 | 0.031683 |
| 5  | 28_enzyme_mimicB1-501EN60a  | 324 | 1613 | 0.030826 | 9.956790  | 2 | 19 | 0.544257 | 1 | 0.030826 |
| 6  | 34_enzyme_mimicB1-501EN59e  | 312 | 1540 | 0.031742 | 9.871795  | 2 | 19 | 0.555165 | 1 | 0.031742 |
| 7  | 17_enzyme_mimicB1-501EN59a  | 313 | 1579 | 0.032338 | 10.089457 | 2 | 19 | 0.540277 | 1 | 0.032338 |
| 8  | 15_enzyme_mimicB1-501EN20e  | 334 | 1622 | 0.029167 | 9.712575  | 4 | 18 | 0.545836 | 1 | 0.029167 |
| 9  | 31_enzyme_mimicB1-501EN321e | 328 | 1655 | 0.030861 | 10.091463 | 2 | 19 | 0.548219 | 1 | 0.030861 |
| 10 | 30_enzyme_mimicB1-501EN259d | 324 | 1582 | 0.030234 | 9.765432  | 3 | 19 | 0.545546 | 1 | 0.030234 |
| 11 | 13_enzyme_mimicB1-501EN60c  | 324 | 1633 | 0.031208 | 10.080247 | 2 | 18 | 0.536101 | 1 | 0.031208 |
| 12 | 19_enzyme_mimicB1-501EN28b  | 334 | 1648 | 0.029634 | 9.868263  | 4 | 19 | 0.541038 | 1 | 0.029634 |
| 13 | 44_enzyme_mimicB1-501EN37d  | 330 | 1666 | 0.030690 | 10.096970 | 2 | 19 | 0.540706 | 1 | 0.030690 |

Protein structures represented through graph-based and biochemical descriptors.

# K-Means Analysis

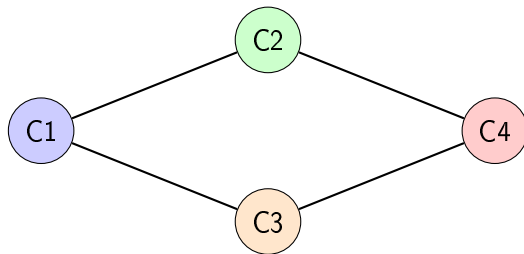
```
k = 2, silhouette = 0.2424  
k = 3, silhouette = 0.2076  
k = 4, silhouette = 0.1974  
k = 5, silhouette = 0.2168  
k = 6, silhouette = 0.2367  
k = 7, silhouette = 0.2680
```

Silhouette analysis used to determine the number of clusters.



2D PCA visualization of clustered enzyme structures.

# Consensus-Based Candidate Selection



- Select representative structures from each cluster.
- Avoid selecting many variants of the same fold.
- Increase structural diversity of experimentally tested candidates.

- Reduce wet-lab validation costs.
- Increase structural diversity of EMs but also preserve catalytic functionality.
- Introduce graph-theoretical and combinatorial descriptors into enzyme design.
- Enable systematic exploration of protein structural space.
- Establish a feedback loop between AI generation and experimental validation.

Towards Reliable and Diverse AI-Generated Enzymes